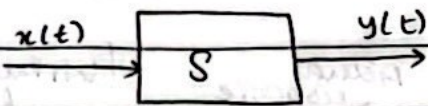


FEEDBACK CONTROL SYSTEM (Mam. Tamania)

Date:

Sun Mon Tue Wed Thu Fri Sat

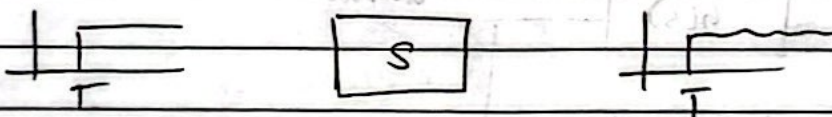
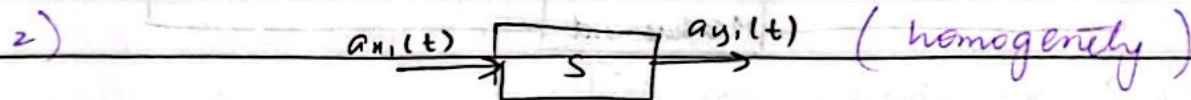
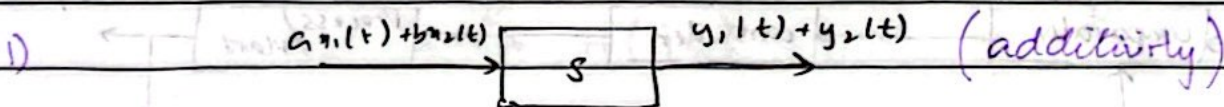
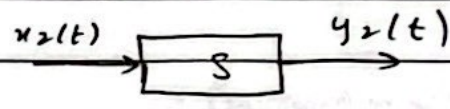
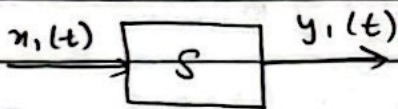
DAY-1



Single input single output
SISO

Linear & Non-linear System:

A system in which superposition theorem is applicable is a linear system



(Time invariance)

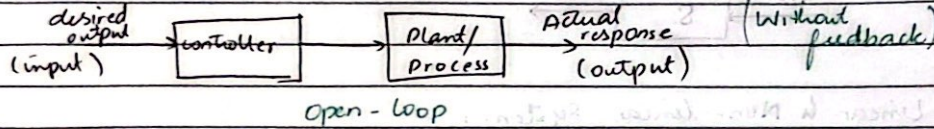
If a system is being checked continuously, it is a feedback system. And if it is also being controlled with being checked, it is feedback control system.

S-YAC

Components of control system:

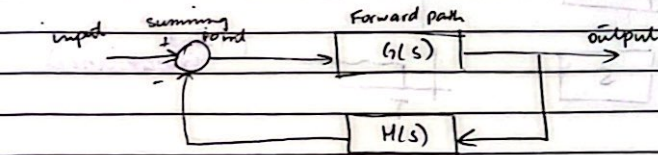
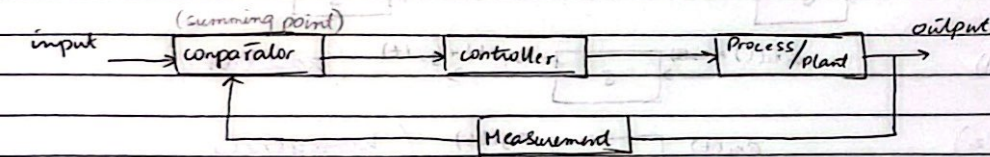
- Input
- Output
- Ways to achieve input & output objective

Control action

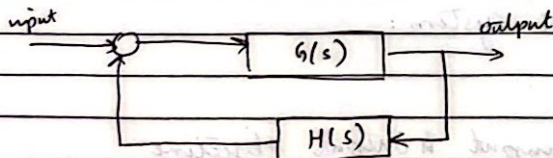


Basic terminologies:

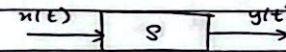
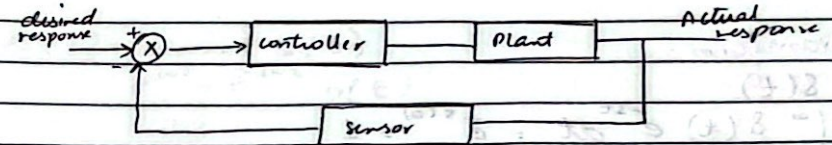
- Plant / Process
- Controller
- input
- output
- Disturbance
- Sensor / Transducer



DAY-2



COPY



$$T.F = \frac{Y(s)}{X(s)}$$

Review of Laplace Transform:

$$\begin{aligned} x(t) &\xrightarrow{S} Y(s) \\ h(t) &\xleftrightarrow{L} H(s) \\ h(t) &\xleftrightarrow{F} H(j\omega) \end{aligned}$$

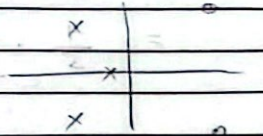
$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad (LT)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (FT)$$

Zero's and pole's of system:

- Poles are the values which are infinity or if we put the denominator = 0. The values will be poles.
- * Zeros are the values which are zero or if we put the numerator = 0. The values will be zero.
- Poles are represented by X
- * Zeros are represented by 0

$$H(s) = \frac{(s-3+4j)(s-3-4j)}{(s+1)(s+2+2j)(s+2-2j)}$$



→ On a LTI system, there must be real and conjugate points part of

→ If the pole is on the left side, it is stable.

Date:

Sun Mon Tue Wed Thu Fri Sat

Laplace Transform:

$$\rightarrow x_1(t) = \delta(t)$$

$$X_1(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-s(0)} = 1$$

$$\text{or } = \int_{-\infty}^{\infty} \delta(t-2) e^{-st} dt = e^{-2s}$$

$$\rightarrow x_2(t) = u(t)$$

$$X_2(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt \Rightarrow \int_0^{\infty} e^{-st} dt$$

$$= \left. \frac{e^{-st}}{-s} \right|_0^{\infty} \Rightarrow -\frac{1}{s} [e^{-\infty} - e^0] \Rightarrow \frac{1}{s}$$

$$\rightarrow x_3(t) = t u(t)$$

$$X_3(s) = \int_{-\infty}^{\infty} t u(t) e^{-st} dt \Rightarrow \int_0^{\infty} t e^{-st} dt$$

$$= \left. t \cdot \frac{e^{-st}}{-s} \right|_0^{\infty} - \int_0^{\infty} 1 \cdot \frac{e^{-st}}{-s} dt$$

$$= \frac{1}{s} \cdot \left. \frac{e^{-st}}{-s} \right|_0^{\infty} \Rightarrow \frac{1}{s^2}$$

$$\rightarrow x_4(t) = e^{-at} u(t)$$

$$X_4(s) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt \Rightarrow \int_0^{\infty} e^{-(s+a)t} dt$$

$$= \left. \frac{e^{-(s+a)t}}{-(s+a)} \right|_0^{\infty} \Rightarrow \frac{1}{s+a}$$

Date:

Sun Mon Tue Wed Thu Fri Sat

$$\rightarrow x_s(t) = \sin \omega t u(t)$$

$$= \frac{e^{j\omega t} - e^{-j\omega t}}{2j} u(t)$$

$$X_s(s) = \frac{1}{2j} \int_0^{\infty} e^{j\omega t} e^{-st} dt - \frac{1}{2j} \int_0^{\infty} e^{-j\omega t} e^{-st} dt$$

$$= \frac{1}{2j} \int_0^{\infty} e^{-(s-j\omega)t} dt - \frac{1}{2j} \int_0^{\infty} e^{-(s+j\omega)t} dt$$

$$= \frac{1}{2j} \left[\frac{e^{-(s-j\omega)t}}{-(s-j\omega)} \right]_0^{\infty} - \frac{1}{2j} \left[\frac{e^{-(s+j\omega)t}}{-(s+j\omega)} \right]_0^{\infty}$$

$$= \frac{1}{2j} \left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right]$$

$$= \frac{\omega}{s^2 + \omega^2}$$

Properties of Laplace Transform:

Linearity:

$$a x_1(t) + b x_2(t) \leftrightarrow a X_1(s) + b X_2(s)$$

Time shifting:

$$x(t-t_0) \leftrightarrow X(s) e^{-st_0}$$

Frequency shifting:

$$e^{-at} x(t) \leftrightarrow X(s+a)$$

Differentiation Property:

$$\frac{dx(t)}{dt} \leftrightarrow sX(s) - x(0)$$

$$\frac{d^2x(t)}{dt^2} \leftrightarrow s^2 X(s) - sx(0) - x'(0)$$

$$\frac{d^n x(t)}{dt^n} \leftrightarrow s^n X(s) - \sum_{k=1}^n s^{n-k} x^{(k)}(0)$$

Date:.....

Sun Mon Tue Wed Thu Fri Sat

Integration Property:

$$\int_0^t x(\tau) d\tau \leftrightarrow \frac{X(s)}{s}$$

DAY-3

Initial value Theorem:

$$f(0^+) = \lim_{s \rightarrow \infty} s F(s)$$

Final value Theorem:

$$f(\infty) = \lim_{s \rightarrow 0} s F(s)$$

$$\frac{d^2 y(t)}{dt^2} + 12 \frac{dy(t)}{dt} + 32 y(t) = 32 x(t)$$

Take Laplace transform

$$s^2 Y(s) + 12s Y(s) + 32 Y(s) = 32 X(s)$$

$$Y(s) = \frac{32 X(s)}{s^2 + 12s + 32}$$

$$X(s) = \frac{1}{s^2 + 12s + 32}$$

$$\frac{X(s)}{Y(s)} = \frac{1}{s^2 + 12s + 32}$$

$$Y(s) = X(s) X(s)$$

$$y(t) = h(t) * x(t)$$

$$x(t) = u(t)$$

$$X(s) = U(s) = \frac{1}{s}$$

Date:.....

Sun Mon Tue Wed Thu Fri Sat

$$1) \quad V(s) = \frac{32}{s(s^2 + 12s + 32)} = \frac{32}{s(s+4)(s+8)}$$

$$= \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+8}$$

$$32 = A(s+4)(s+8) + B s(s+8) + C s(s+4)$$

$$\text{put } s = -4$$

$$32 = -16B$$

$$B = -2$$

$$\text{put } s = -8$$

$$32 = 32C$$

$$C = 1$$

$$\text{put } s = 0$$

$$A = 1$$

$$Y(s) = \frac{1}{s} - \frac{2}{s+4} + \frac{1}{s+8}$$

$$y(t) = u(t) - 2e^{-4t}u(t) + e^{-8t}u(t)$$

$$= u(t) [1 - 2e^{-4t} + e^{-8t}]$$

$$2) \quad F(s) = \frac{2}{(s+1)(s+2)^2}$$

$$= \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

$$2 = A(s+2)^2 + B(s+1)(s+2) + C(s+1)$$

$$\text{put } s = -1$$

$$2 = A$$

$$A = 2$$

put $s = -2$
 $2 = -C(8+2)(1+2) \dots$
 $C = -2$

put $s = 0$
 $2 = 2(2)^2 + B(2) + (-2)(1) \dots$
 $B = 2 - 8 + 2$
 $B = -2$

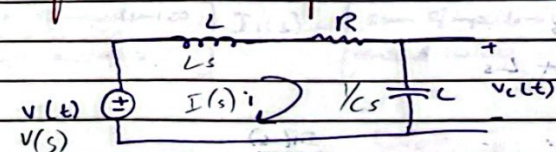
$F(s) = \frac{2}{s+1} + \frac{-2}{s+2} - \frac{2}{(s+2)^2}$
 $= 2e^{-t}u(t) - 2e^{-2t}u(t) - 2te^{-2t}u(t)$
 $= 2u(t) [e^{-t} - e^{-2t} - te^{-2t}]$

3) $F(s) = \frac{3}{s(s^2+2s+5)}$
 $= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$
 $3 = A(s^2+2s+5) + B(s)(s+2) + C(s)(s+1)$
 $s = 0 \Rightarrow 3 = 5A \Rightarrow A = \frac{3}{5}$
 $A + B = 0 \Rightarrow B = -\frac{3}{5}$
 $2A + C = 0 \Rightarrow C = -\frac{6}{5}$

$= \frac{3}{5} - \frac{3}{5} e^{-t} \sqrt{1+(1/2)^2} \left[\cos 2t + \frac{1}{2} \sin 2t \right] u(t)$
 $= 0.6 - 0.67 e^{-t} \left[\cos 2t + \frac{1}{2} \sin 2t \right] u(t)$
 $= [0.6 - 0.67 e^{-t} \cos(2t - \phi)] u(t)$
 $\phi = \tan^{-1} \frac{1}{2}$

4) $G(s) = \frac{2s+1}{s^2+2s+6} = \frac{Y(s)}{X(s)}$
 $(s^2+2s+6)Y(s) = (2s+1)X(s)$
 $s^2 Y(s) + 2s Y(s) + 6Y(s) = 2s X(s) + X(s)$
 $\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 6y(t) = 2 \frac{dx(t)}{dt} + x(t)$

Transfer Function of electrical Network



Find $\frac{V_C(s)}{V(s)}$
 $-V(t) + L \frac{di}{dt} + Ri + V_C(t) = 0$
 $-V(t) + L \frac{d^2 V_C(t)}{dt^2} + (R + LC) \frac{dV_C(t)}{dt} + V_C(t) = 0$
 $LC s^2 V_C(s) + (R + LC) s V_C(s) + V_C(s) = V(s)$
 $\frac{V_C(s)}{V(s)} = \frac{1}{LC s^2 + (R + LC)s + 1}$

$$= 1/LC$$

$$s \rightarrow R/Ls + 1/LC$$

Another Method: (2nd Method)

$$-V(s) + Ls I(s) + RI(s) + I(s) \frac{1}{Cs} = 0$$

$$\left(Ls + R + \frac{1}{Cs} \right) I(s) = V(s)$$

$$\left(Ls + R + \frac{1}{Cs} \right) (Cs) V_c(s) = V(s)$$

$$\{ LCs^2 + RCs + 1 \} V_c(s) = V(s)$$

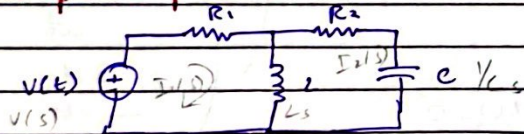
$$\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}$$

3rd Method:

Apply VDR in s-domain

$$V_c(s) = V(s) \frac{1/Cs}{1/Cs + R + Ls}$$

Multiple loop circuit:



$$\frac{I_1(s)}{V(s)}$$

$$i_2 = C \frac{dv_c}{dt}$$

$$-V(s) + RI_1(s) + Ls(I_1(s) - I_2(s)) = 0 \rightarrow (1)$$

$$R_2 I_2(s) + \frac{1}{Cs} I_2(s) + Ls(I_2(s) - I_1(s)) = 0 \rightarrow (2)$$

$$(Ls + R_1) I_1(s) - Ls I_2(s) = V(s) \text{ simplifying (1)}$$

$$-Ls I_1(s) + (Ls + R_2) I_2(s) = 0 \text{ simplifying (2)}$$

$$\begin{bmatrix} Ls + R_1 & -Ls \\ -Ls & 1/Cs + Ls + R_2 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V(s) \\ 0 \end{bmatrix}$$

$$I_1(s) = \frac{\begin{vmatrix} V(s) & -Ls \\ 0 & R_2 + Ls + 1/Cs \end{vmatrix}}{\Delta}$$

$$0 = \begin{vmatrix} Ls + R_1 & -Ls \\ -Ls & R_2 + Ls + 1/Cs \end{vmatrix}$$

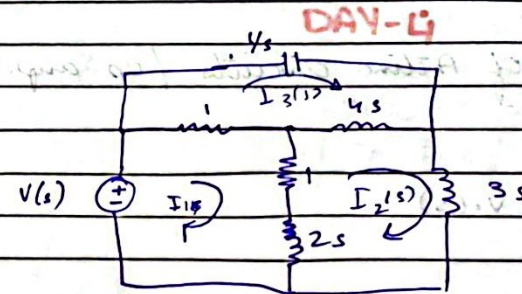
$$= (Ls + R_1)(R_2 + 1/Cs + Ls) - L^2 s^2$$

$$I_2(s) = \frac{\begin{vmatrix} Ls + R_1 & V(s) \\ -Ls & 0 \end{vmatrix}}{\Delta}$$

Direct Method: (short cut)

$$\left[\begin{matrix} \text{sum of impedances} \\ \text{around Mesh 1} \end{matrix} \right] I_1(s) - \left[\begin{matrix} \text{sum of impedances} \\ \text{common to mesh 1 \& mesh 2} \end{matrix} \right] I_2(s) = \left[\begin{matrix} \text{sum of applied} \\ \text{voltages at Mesh 1} \end{matrix} \right]$$

$$- \left[\begin{matrix} \text{sum of impedances} \\ \text{common to mesh 1 \& mesh 2} \end{matrix} \right] I_1(s) + \left[\begin{matrix} \text{sum of impedances} \\ \text{around mesh 2} \end{matrix} \right] I_2(s) = \left[\begin{matrix} \text{sum of applied} \\ \text{voltages at mesh 2} \end{matrix} \right]$$



$$-V_s + (I_1 - I_2) + (I_1 - I_2) + (I_1 - I_2) 2s = 0 \rightarrow (1)$$

$$2s(I_1 - I_2) + (I_2 - I_1) + 4s(I_2 - I_1) + 3s I_2 = 0$$

$$4s + \frac{1}{s} + 1$$

$$4s^2 + 1 + s \quad 4s^2 + s + 1$$

Date:

Sun Mon Tue Wed Thu Fri Sat

$$(2+2s)(I_1(s)) - (1+2s)(I_2(s)) - 1(I_3(s)) = V_i$$

Mesh 1:

$$(2+2s)I_1(s) - (1+2s)I_2(s) - I_3(s) = V_i$$

Mesh 2:

$$-(1+2s)I_1(s) + (9s+1)I_2(s) - 4sI_3(s) = 0$$

Mesh 3:

$$-I_1(s) - 4sI_2(s) + (4s^2+s+1)I_3(s) = 0$$

$$\begin{bmatrix} 2+2s & -(1+2s) & -1 \\ -1+2s & 9s+1 & -4s \\ -1 & -4s & 4s^2+s+1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} V_i(s) \\ 0 \\ 0 \end{bmatrix}$$

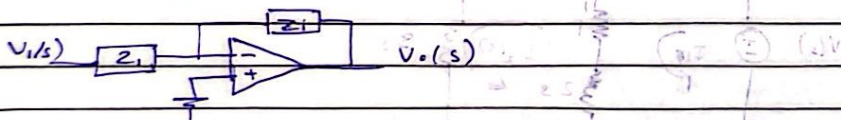
$$I_1(s) = \frac{V_i(s) - (1+2s)I_2(s) - I_3(s)}{2+2s}$$

$$\Delta = (2+2s)[(9s+1)(4s^2+s+1) - 4s(-1+2s)] + (1+2s)[-1(4s^2+s+1) - 4s(-1)] - 1[4s(1+2s) + 9s+1]$$

2-VAO

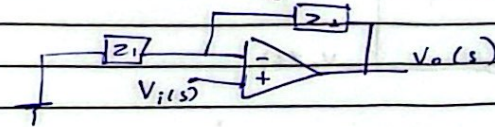
Transfer Function of Active circuits / op-amp based ccts.

→ Inverting:



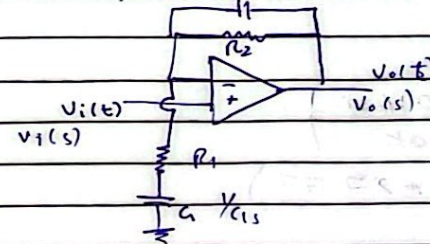
$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1}$$

→ Non Inverting



$$\frac{V_o(s)}{V_i(s)} = \frac{1+Z_2}{Z_1}$$

Example:



$$Z_2 = R_2 \parallel \frac{1}{C_2 s} = \frac{R_2}{1+R_2 C_2 s}$$

$$Z_1 = R_1 + \frac{1}{C_1 s} = \frac{1+R_1 C_1 s}{C_1 s}$$

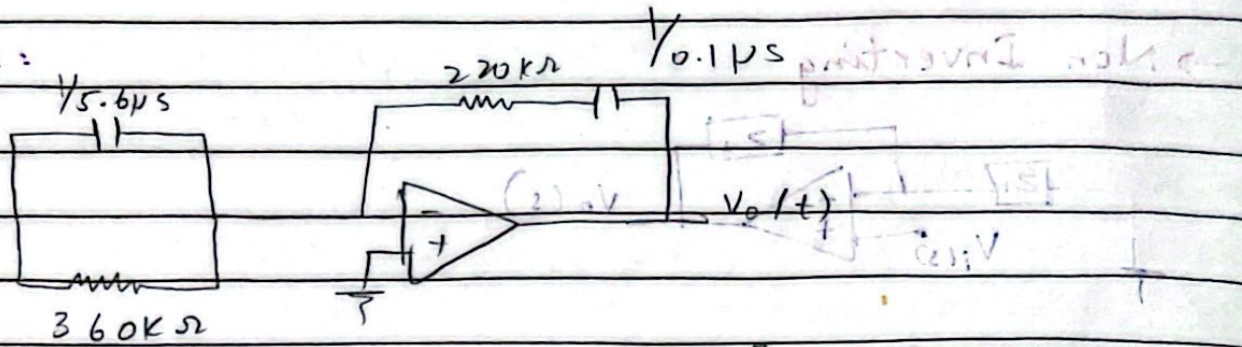
$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1} = -\frac{R_2 C_1 s}{1+R_1 C_1 s + R_2 C_2 s + R_1 R_2 C_1 C_2 s^2}$$

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2}{Z_1}$$

$$\begin{aligned} &= 1 + \frac{R_2}{1+R_2 C_2 s} \cdot \frac{1+R_1 C_1 s}{1+R_1 C_1 s} \\ &= \frac{(1+R_1 C_1 s)(1+R_2 C_2 s) + R_2 C_1 s}{(1+R_1 C_1 s)(1+R_2 C_2 s)} \\ &= \frac{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1)s + 1}{(1+R_1 C_1 s)(1+R_2 C_2 s)} \end{aligned}$$

there are 2 poles → from denominator
there are 2 zeros → from numerator
It is of second order → from poles

Example:



$$Z_2 = 220K + \frac{1}{0.1\mu s} = 220K + 10^7 = 220Ks + 10^7$$

$$Z_1 = \frac{360K \left(\frac{1}{5.6\mu s} \right)}{360K + \frac{1}{5.6\mu s}} = 360K \frac{1 + (360K)(5.6\mu)s}{1 + 2.016s}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{220K + 10^7/s}{360K / (1 + 2.016s)}$$

$$= \frac{(220Ks + 10^7) \times (1 + 2.016s)}{360K}$$

$$= \frac{(220Ks + 10^7)(1 + 2.016s)}{360K}$$

$$= \frac{(220Ks + 10^7)(1 + 2.016s)}{360K}$$

$$= \frac{(220Ks + 10^7)(1 + 2.016s)}{360K}$$

$$= \frac{(220Ks + 10^7)(1 + 2.016s)}{360K}$$

$$= \frac{(220Ks + 10^7)(1 + 2.016s)}{360K}$$

$$= \frac{(220Ks + 10^7)(1 + 2.016s)}{360K}$$

$$\frac{X(s)}{n(t)} \rightarrow \frac{dn(t)}{dt} \quad \text{(differentiation)}$$

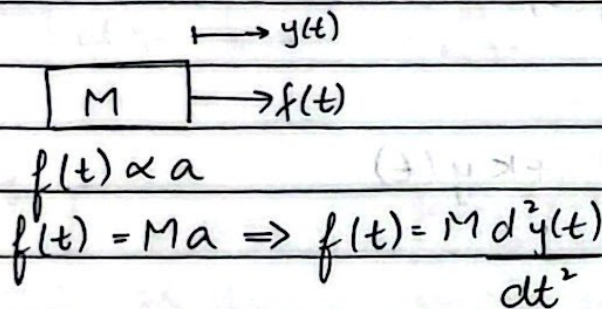
division $\rightarrow$ integration

DAY - 5**Transfer Function of Mechanical Systems :**

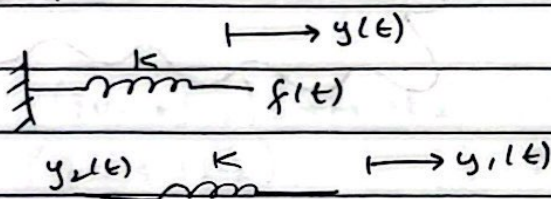
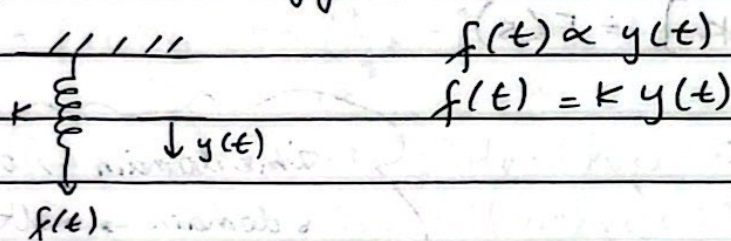
- Translational Mechanical Systems
- Mass, Spring, Damper
 - Force, displacement

Mass:

- stores kinetic energy
- rigid bodies

**Spring:**

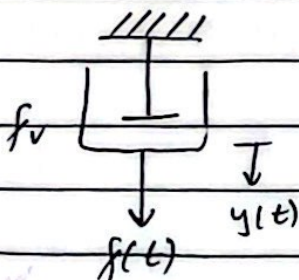
- stores potential energy



$$f(t) = K (y_1(t) - y_2(t))$$

Damper:

- Frictional element



$$f(t) \propto v(t)$$

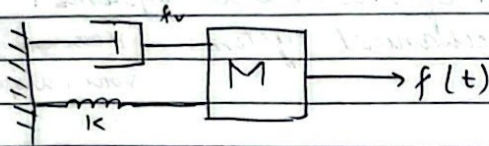
$$f(t) \propto \frac{dy(t)}{dt} \Rightarrow f(t) = f_v \frac{dy(t)}{dt}$$

→ f_v = viscous friction coefficient

2-PAQ

Ideal Spring Mass Damper System:

Question



$$f(t) = M \frac{d^2 y}{dt^2} + f_v \frac{dy}{dt} + K y(t)$$

$$Y(s) = ?$$

$$F(s)$$

$$F(s) = M s^2 Y(s) + f_v s Y(s) + K Y(s)$$

$$Y(s) [M s^2 + f_v s + K] = F(s)$$

$$Y(s) = \frac{1}{M s^2 + f_v s + K}$$

$$F(s) \quad M s^2 + f_v s + K$$

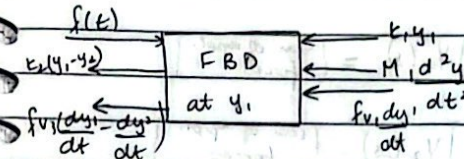
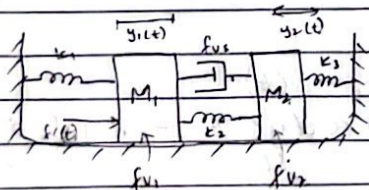
$$M = 3, f_v = 2, K = 1$$

$$Y(s) = \frac{1}{3s^2 + 2s + 1}$$

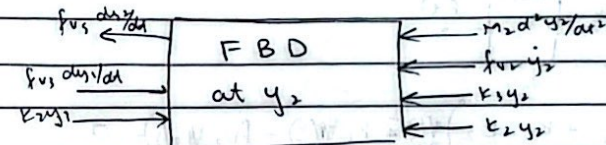
$$F(s) \quad 3s^2 + 2s + 1$$

time domain $\rightarrow$ convolution
s domain $\rightarrow$ multiplication

Question



$$f(t) = M_1 \frac{d^2 y_1}{dt^2} + f_{v1} \frac{dy_1}{dt} + f_{v2} \left[\frac{dy_1}{dt} - \frac{dy_2}{dt} \right] + K_1 y_1 + K_2 (y_1 - y_2) \rightarrow (1)$$



$$K_2 y_1 + f_{v2} \frac{dy_1}{dt} = M_2 \frac{d^2 y_2}{dt^2} + f_{v2} \frac{dy_2}{dt} + K_3 y_2 + K_2 y_2 + f_{v1} \frac{dy_2}{dt} \rightarrow (2)$$

$$[M_1 s^2 + (f_{v1} + f_{v2})s + K_1 + K_2] Y_1(s) - (f_{v2}s + K_2) Y_2(s) = F(s)$$

$$-(f_{v2}s + K_2) Y_1(s) + [M_2 s^2 + (f_{v2} + f_{v1})s + K_2 + K_3] Y_2(s) = 0$$

$$\begin{bmatrix} M_1 s^2 + (f_{v1} + f_{v2})s + K_1 + K_2 & -(f_{v2}s + K_2) \\ -(f_{v2}s + K_2) & M_2 s^2 + (f_{v2} + f_{v1})s + K_2 + K_3 \end{bmatrix} \begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$F(t) = M \frac{d^2 y(t)}{dt^2} \leftrightarrow F(s) = M s^2 Y(s) \Rightarrow F(s)/Y(s) = M s^2$$

$$F(t) = f_v \frac{dy(t)}{dt} \leftrightarrow F(s) = f_v s Y(s) \Rightarrow F(s)/Y(s) = f_v s$$

$$F(t) = K y(t) \leftrightarrow F(s) = K Y(s) \Rightarrow F(s)/Y(s) = K$$

Date:

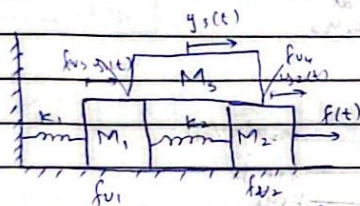
Sun Mon Tue Wed Thu Fri Sat

$$\left[\begin{array}{c} \text{sum of imp. connected} \\ \text{at point of motion } y_1 \end{array} \right] Y_1(s) - \left[\begin{array}{c} \text{common b/w} \\ y_1 \text{ \& } y_2 \end{array} \right] Y_2(s) = \left[\begin{array}{c} \text{sum of appl.} \\ \text{forces} \end{array} \right]$$

$$- \left[\begin{array}{c} \text{common b/w} \\ y_1 \text{ \& } y_2 \end{array} \right] Y_1(s) + \left[\begin{array}{c} \text{sum of imp. connected} \\ \text{at point of motion } y_2 \end{array} \right] Y_2(s) = \left[\begin{array}{c} \text{sum of appl.} \\ \text{forces} \end{array} \right]$$

DAY-6

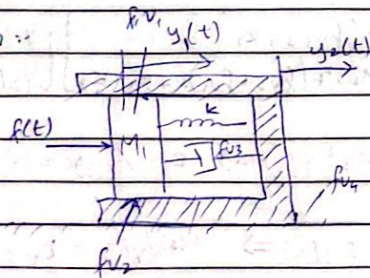
Question:



$$\begin{aligned} (M_1 s^2 + f_{v1} s + f_{v2} s + K_1 + K_2) Y_1(s) - K_2 Y_2(s) - f_{v2} Y_3(s) &= 0 \\ -K_2 Y_1(s) + (M_2 s^2 + f_{v2} s + f_{v3} s + K_2) Y_2(s) - f_{v3} Y_3(s) &= F(s) \\ -f_{v3} Y_1(s) - K_2 Y_2(s) + (M_3 s^2 + f_{v3} s + f_{v4}) Y_3(s) &= 0 \end{aligned}$$

$$M_1 \frac{d^2 y_1}{dt^2} + f_{v1} \frac{dy_1}{dt} + f_{v2} \frac{dy_1}{dt} + K_1 y_1 + K_2 y_2 - K_2 y_2 - f_{v2} \frac{dy_3}{dt} = 0$$

Question:



Take $f_{v4} = f_{v3} = f_{v2} = f_{v1} = 1 \text{ Ns/m}$
 $M_1 = M_2 = 1 \text{ kg}$

Date:

Sun Mon Tue Wed Thu Fri Sat

$$K = 1 \text{ N/m}$$

Find $Y_2(s)$
 $F(s)$

$$\begin{aligned} (M_1 s^2 + f_{v1} s + f_{v2} s + f_{v3} s + K) Y_1(s) - (F_{v1} + F_{v2} + F_{v3} + K) Y_2(s) \\ - (F_{v1} + F_{v2} + F_{v3} + K) Y_1(s) + (M_2 s^2 + (F_{v1} + F_{v2} + F_{v3} + F_{v4}) s + K) Y_2(s) = F(s) \end{aligned}$$

$$\begin{bmatrix} s^2 + 3s + 1 & -(3s + 1) \\ -(3s + 1) & s^2 + 4s + 1 \end{bmatrix} \begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$Y_2(s) = \frac{\begin{vmatrix} s^2 + 3s + 1 & F(s) \\ -(3s + 1) & 0 \end{vmatrix}}{\Delta}$$

$$\begin{aligned} \Delta &= s^4 + 7s^3 + 5s^2 + s \\ &= s(s^3 + 7s^2 + 5s + 1) \end{aligned}$$

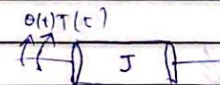
$$Y_2(s) = \frac{(3s + 1) F(s)}{\Delta}$$

$$\frac{Y_2(s)}{F(s)} = \frac{3s + 1}{s(s^3 + 7s^2 + 5s + 1)}$$

Rotational Mechanical Systems:

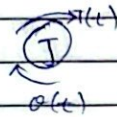
→ Torque $T(t)$ → Angular displacement $\theta(t)$

Inertial Mass:



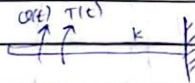
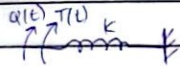
$$T(t) \propto \frac{d^2 \theta(t)}{dt^2}$$

$$T(t) = J \frac{d^2 \theta(t)}{dt^2}$$



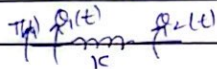
J = Moment of inertia

Torsional Spring:



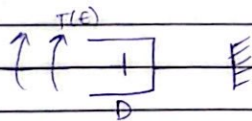
$$T(t) \propto \theta(t)$$

$$T(t) = K \theta(t)$$



$$T(t) = K(\theta_1(t) - \theta_2(t))$$

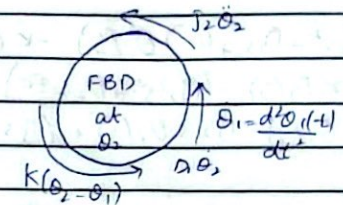
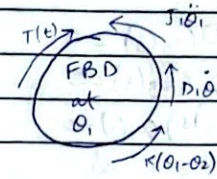
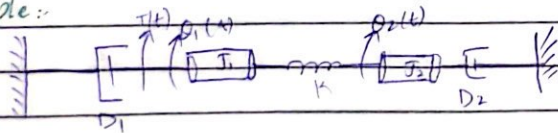
Torsional damper:



$$T(t) \propto \frac{d\theta(t)}{dt}$$

$$T(t) = D \frac{d\theta(t)}{dt}$$

Example:



For θ_1 :

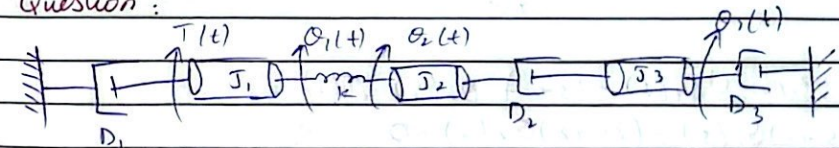
$$J_1 \frac{d^2 \theta_1}{dt^2} + D_1 \frac{d\theta_1}{dt} + K(\theta_1 - \theta_2) = T(t)$$

$$J_2 \frac{d^2 \theta_2}{dt^2} + D_2 \frac{d\theta_2}{dt} + K(\theta_2 - \theta_1) = 0$$

$$(J_1 s^2 + D_1 s + K)\theta_1(s) - K\theta_2(s) = T(s)$$

$$-K\theta_1(s) + (J_2 s^2 + D_2 s + K)\theta_2(s) = 0$$

Question:



Find $\theta_3(s)$
 $T(s)$

$$J_1 \frac{d^2 \theta_1}{dt^2} + D_1 \frac{d\theta_1}{dt} + K(\theta_1 - \theta_2) = T(t)$$

$$J_2 \frac{d^2 \theta_2}{dt^2} + D_2 \left(\frac{d\theta_2}{dt} - \frac{d\theta_3}{dt} \right) + K(\theta_2 - \theta_1) = 0$$

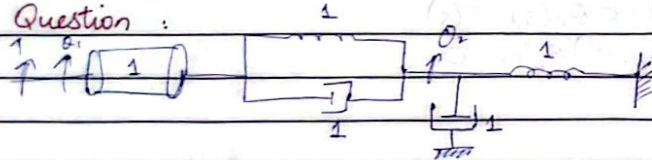
$$J_3 \frac{d^2 \theta_3}{dt^2} + D_2 \left(\frac{d\theta_3}{dt} - \frac{d\theta_2}{dt} \right) + D_3 \frac{d\theta_3}{dt} = 0$$



$$\begin{aligned}
 (J_1 s^2 + D_1 s + K) \theta_1(s) - K \theta_2(s) - 0 \theta_3(s) &= T s \\
 -K \theta_1(s) + (J_2 s^2 + D_2 s + K) \theta_2(s) - D_2 s \theta_3(s) &= 0 \\
 0 \theta_1(s) - D_2 s \theta_2(s) + (J_3 s^2 + (D_2 + D_3) s) \theta_3(s) &= 0
 \end{aligned}$$

$$\begin{bmatrix} J_1 s^2 + D_1 s + K & -K & 0 \\ -K & J_2 s^2 + D_2 s + K & -D_2 s \\ 0 & -D_2 s & J_3 s^2 + (D_2 + D_3) s \end{bmatrix} \begin{bmatrix} \theta_1(s) \\ \theta_2(s) \\ \theta_3(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \end{bmatrix}$$

Question:



$$\begin{aligned}
 (s^2 + s + 1) \theta_1(s) - (s + 1) \theta_2(s) &= T(s) \\
 -(s + 1) \theta_1(s) + (2s + 2) \theta_2(s) &= 0
 \end{aligned}$$

$$\begin{bmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & 2s + 2 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

$$\theta_2(s) = \frac{s^2 + s + 1}{-(s + 1)} \cdot \frac{T(s)}{0}$$

$$\Delta = \begin{vmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & 2s + 2 \end{vmatrix}$$

DAY-7

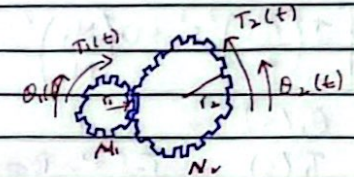
Transfer Function of system with gears:

→ used to transfer energy from one part of the system to other.

→ Provide a trade off between speed & torque.

→ To mathematically model gears, we use gear ratio

$$\frac{\omega_1}{\omega_2} = \frac{N_1}{N_2}$$



→ Work done by input gear is same as the work done by output gear.

→ Distance travelled along the circumference of both gears is same.

$$S = \omega \theta$$

$$\omega_1 \theta_1 = \omega_2 \theta_2$$

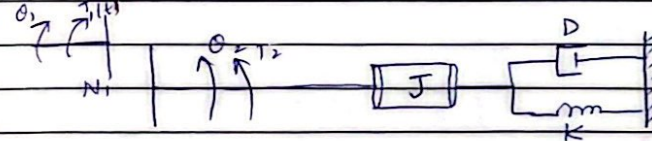
$$T_1 \theta_1 = T_2 \theta_2$$

$$\frac{\omega_1}{\omega_2} = \frac{\theta_2}{\theta_1}$$

$$\frac{T_1}{T_2} = \frac{\theta_2}{\theta_1}$$

$$\frac{\omega_1}{\omega_2} = \frac{T_1}{T_2} = \frac{N_1}{N_2} = \frac{\theta_2}{\theta_1}$$

Example:

Find $\frac{\theta_2(s)}{T_1(s)}$

F-VAD

$$(Js^2 + Ds + K) \theta_2(s) = T_2(s)$$

$$(Js^2 + Ds + K) \left(\frac{N_1}{N_2} \right) \theta_1(s) = \frac{N_2}{N_1} T_1(s)$$

$$\left[J \left(\frac{N_1}{N_2} \right)^2 s^2 + D \left(\frac{N_1}{N_2} \right)^2 s + K \left(\frac{N_1}{N_2} \right)^2 \right] \theta_1(s) = T_1(s)$$

$$\theta_1(s) = \frac{1}{T_1(s) \left[J_{eq} s^2 + D_{eq} s + K_{eq} \right]}$$

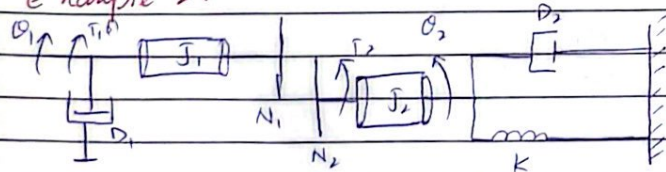
$$J_{eq} = J \left(\frac{N_1}{N_2} \right)^2$$

$$D_{eq} = D \left(\frac{N_1}{N_2} \right)^2$$

$$K_{eq} = K \left(\frac{N_1}{N_2} \right)^2$$

$$\left(\frac{\text{No. of teeth at destination}}{\text{No. of teeth at source}} \right)^2$$

Example 2:



$$\text{Find } \theta_2(s) ?$$

$$T_2(s)$$

At θ_2

$$J_{eq} = J_2 + J_1 \left(\frac{N_2}{N_1} \right)^2$$

$$D_{eq} = D_2 + D_1 \left(\frac{N_2}{N_1} \right)^2$$

$$\frac{T_2}{T_1} = \frac{N_2}{N_1}$$

$$T_2 = \frac{N_2}{N_1} T_1$$

$$\frac{\theta_2}{\theta_1} = \frac{N_1}{N_2}$$

$$\theta_2 = \frac{N_1}{N_2} \theta_1$$

$$K_{eq} = K$$

$$\frac{\theta_2(s)}{T_2(s)} = \frac{1}{J_{eq} s^2 + D_{eq} s + K_{eq}}$$

$$\theta_2(s) ?$$

$$T_1(s)$$

$$(J_{eq} s^2 + D_{eq} s + K_{eq}) \theta_2(s) = \frac{N_2}{N_1} T_1(s)$$

$$(J_{eq} s^2 + D_{eq} s + K_{eq}) \theta_2(s) = \frac{N_2}{N_1} T_1(s)$$

$$\theta_2(s) = \frac{N_2/N_1}{T_1(s) [J_{eq} s^2 + D_{eq} s + K_{eq}]}$$

$$\text{Find } \frac{\theta_2(s)}{T_1(s)}$$

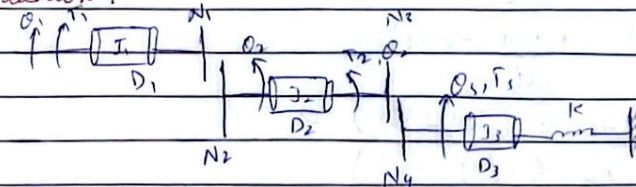
$$(J_{eq} s^2 + D_{eq} s + K_{eq}) \theta_2(s) = T_1(s)$$

$$J_{eq} = J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2$$

$$D_{eq} = D_1 + D_2 \left(\frac{N_1}{N_2} \right)^2$$

$$K_{eq} = K \left(\frac{N_1}{N_2} \right)^2$$

Question:

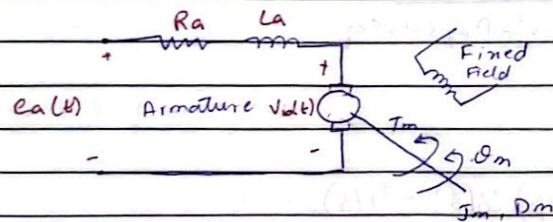


Find $\frac{\Theta_m(s)}{T_m(s)}$

$$J_T = \left[J_3 + J_2 \left(\frac{N_4}{N_5} \right)^2 + J_1 \left(\frac{N_2}{N_1} \right)^2 \left(\frac{N_4}{N_5} \right)^2 \right] s^2$$

DAY-8

Transfer Function of electromechanical system (DC servomotor)



$$V_b(t) \propto \frac{d\Theta_m}{dt}$$

$$V_b(t) = K_b \frac{d\Theta_m}{dt} \rightarrow \textcircled{1}$$

$$T_m(t) \propto I_a(t)$$

$$T_m(t) = K_t I_a(t)$$

$$T_m(s) = K_t I_a(s) \rightarrow \textcircled{2}$$

$$-E_a(t) + R_a I_a(t) + L_a \frac{dI_a}{dt} + V_b(t) = 0$$

$$-E_a(s) + R_a I_a(s) + L_a s I_a(s) + V_b(s) = 0$$

$$(R_a + L_a s) I_a(s) + K_b s \Theta_m(s) = E_a(s)$$

$$\frac{(R_a + L_a s) T_m(s)}{K_t} + K_b s \Theta_m(s) = E_a(s)$$

$$\frac{J_m s^2 \Theta_m}{dt^2} + \frac{D_m s \Theta_m}{dt} = T_m(t)$$

$$(J_m s^2 + D_m s) \Theta_m(s) = T_m(s)$$

$$(R_a + L_a s) (J_m s^2 + D_m s) \Theta_m(s) + K_b s \Theta_m(s) = E_a(s)$$

$$\frac{K_t}{K_t} \left[(R_a + L_a s) (J_m s^2 + D_m s) + K_b s \right] \Theta_m(s) = E_a(s)$$

$$\frac{\Theta_m(s)}{E_a(s)} = \frac{K_t}{(R_a + L_a s) (J_m s^2 + D_m s) + K_b s}$$

$$= \frac{K_t / R_a J_m}{s^2 + \frac{D_m}{J_m} s + \frac{K_b K_t}{R_a J_m}}$$

$$= \frac{K}{s(s + \alpha)} \quad \text{where } \alpha = \frac{D_m + K_b K_t}{J_m R_a}$$

State Space Representation :

input, output

state variables

$$x_1(t), x_2(t), x_3(t)$$

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

A → state matrix

B → input matrix

C → output matrix

D → Direct transmission matrix

- How many state variable (C & L) (min 2)
- How to assign
- Write state equations $\rightarrow$ should be first order differential eq.
- Write output equations $\rightarrow$ state space model

Output relation:

linear algebraic eq. also show relation ship b/w state variable & algebraic variable.

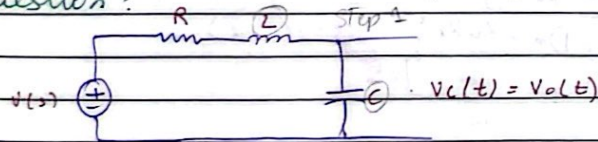
n = state variable

p = output variables

m = input variable

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \\ \\ \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Question:



Let

$$x_1 = V_c(t)$$

$$x_2 = i_L(t)$$

} step 2

$$i_L(t) = C \frac{dV_c(t)}{dt}$$

$$x_2 = C \frac{dx_1}{dt} \Rightarrow \frac{dx_1}{dt} = \frac{1}{C} x_2$$

$$-V(t) + R i_L(t) + L \frac{di_L}{dt} + V_c(t) = 0$$

$$\frac{di_L}{dt} = -\frac{1}{L} V_c(t) - \frac{R}{L} i_L(t) + \frac{1}{L} V(t)$$

$$\frac{dx_2}{dt} = -\frac{1}{L} x_1 - \frac{R}{L} x_2 + \frac{1}{L} V(t)$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} 0 & 1/C \\ -1/L & -R/L \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} V(t)$$

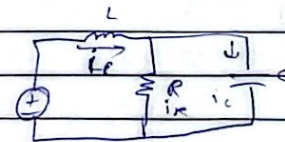
$$y = [1 \ 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{Let } V_R(t) = V_0$$

$$V_R(t) = i_L(t) R$$

$$y = R \times L \begin{bmatrix} 0 & R \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Question:



$$\begin{aligned} V_c &= x_1 \\ i_L &= x_2 \\ i_L &= C \frac{dV_c}{dt} \rightarrow x_1 \end{aligned}$$

$$\frac{dx_1}{dt} = \frac{1}{C} x_2$$

DAV-9

$$V_L = L \frac{di_L}{dt} \quad \text{Let}$$

$$V_C = x_1 ; i_L = x_2 \quad \therefore i_C = i_R + i_L$$

$$\frac{dx_2}{dt} = \frac{1}{L} V_L \quad i_C = C \frac{dV_C}{dt} \Rightarrow i_C = C \frac{dx_1}{dt} \Rightarrow \frac{dx_1}{dt} = \frac{1}{C} i_C = \frac{1}{C} (i_C - i_L)$$

$$= \frac{1}{C} (i_L - \frac{V_L}{R})$$

$$= \frac{1}{C} x_2 - \frac{1}{RC} x_1 \Rightarrow \frac{dx_1}{dt} = -\frac{1}{RC} x_1 + \frac{1}{C} x_2 \rightarrow (1)$$

$$V_L = L \frac{di_L}{dt} \Rightarrow \frac{dx_2}{dt} = \frac{1}{L} V_L \rightarrow (2)$$

$$-V(t) + V_L + V_R = 0$$

$$L \frac{dx_2}{dt} = V(t) - V_C$$

$$\frac{dx_2}{dt} = \frac{1}{L} (-V_C + V(t))$$

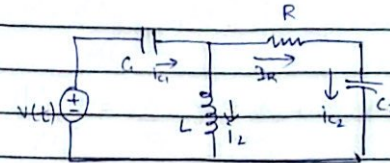
$$\frac{dx_1}{dt} = -\frac{1}{RC} x_1 + \frac{1}{C} V(t) \rightarrow (3)$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} -1/RC & 1/C \\ -1/L & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} V(t)$$

$$y = \begin{bmatrix} -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + V(t)$$

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

Obtain State Space Model. Take V_C as output.

$$V_{C1} = x_1$$

$$i_L = x_2$$

$$V_C = x_3$$

$$i_{C1} = C_1 \frac{dV_{C1}}{dt} \Rightarrow \frac{dx_1}{dt} = i_{C1} \rightarrow (1)$$

$$V_L = L \frac{di_L}{dt} \Rightarrow \frac{dx_2}{dt} = \frac{1}{L} V_L \rightarrow (2)$$

$$i_{C2} = C_2 \frac{dV_C}{dt} \Rightarrow \frac{dV_C}{dt} = \frac{1}{C_2} i_{C2}$$

$$V(t) = C_1 \frac{dV_{C1}}{dt} + L \frac{di_L}{dt} + V_C$$

$$= \frac{C_1}{C_1} \frac{dx_1}{dt} + \frac{L}{L} \frac{dx_2}{dt} + x_3$$

$$-V(t) + V_{C1} + V_C$$

$$V(t) = -V_{C1} + V(t) + V_C$$

put in eq (1)

$$= \frac{1}{L} (-V_{C1} + V(t))$$

$$\frac{dx_2}{dt} = -\frac{1}{L} x_1 + \frac{1}{L} V(t)$$

P-VAD

$$i_C = i_L + i_R$$

$$\frac{dx_1}{dt} = \frac{1}{C_1} (i_L + i_R)$$

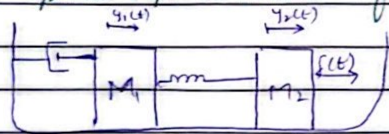
$$= \frac{1}{C_1} \left(x_2 + \frac{V_L - V_C}{R} \right)$$

$$\frac{dx_1}{dt} = \frac{1}{C_1} \left(R x_2 + x_2 - x_1 + V(t) \right)$$

$$= \frac{1}{C_1} \left(-x_1 + R x_2 + x_2 + V(t) \right)$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} 1/R C_1 & 1/C_1 & -1/R C_1 \\ -1/L_1 & 0 & 0 \\ -1/R L_1 & 0 & 1/R L_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1/R C_1 \\ 1/L_1 \\ 1/R C_1 \end{bmatrix} V(t)$$

State space representation of Mechanical systems:



are obtained from eq. or point of motion or displacement rather than energy storage element. Take displacement & velocity at each point as state variable.

The no. of state eq. for each eq. is the order of the equation.

$$y_1 = x_1$$

$$\frac{dy_1}{dt} = x_2 \rightarrow \frac{dx_1}{dt} = x_2$$

$$y_2 = x_3$$

$$\frac{dy_2}{dt} = x_4 \rightarrow \frac{dx_3}{dt} = x_4$$

$$M_1 \frac{d^2 y_1}{dt^2} + f_v \frac{dy_1}{dt} + k(y_1 - y_2) = 0 \rightarrow \text{①}$$

$$M_2 \frac{d^2 y_2}{dt^2} + k(y_2 - y_1) - f(t) = 0 \rightarrow \text{②}$$

$$\frac{d^2 y_1}{dt^2} = \frac{k}{M_1} y_1 + \frac{k}{M_1} y_2 - \frac{f_v}{M_1} \frac{dy_1}{dt}$$

$$\frac{dx_2}{dt} = -\frac{k}{M_1} x_1 - \frac{f_v}{M_1} x_2 + \frac{k}{M_1} x_3$$

$$\frac{d^2 y_2}{dt^2} = \frac{k}{M_2} y_1 - \frac{k}{M_2} y_2 + \frac{1}{M_2} f(t)$$

$$\frac{dx_4}{dt} = \frac{k}{M_2} x_1 - \frac{k}{M_2} x_3 + \frac{1}{M_2} f(t)$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \\ dx_4/dt \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -k/M_1 & -f_v/M_1 & k/M_1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/M_2 & 0 & -k/M_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M_2 \end{bmatrix} f(t)$$

$$y_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

01-VAD

(8)

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

$$C(s) \cdot [s^3 + 9s^2 + 26s + 24] = 24R(s)$$

$$\frac{d^3 c(t)}{dt^3} + 9 \frac{d^2 c(t)}{dt^2} + 26 \frac{dc(t)}{dt} + 24c(t) = 24r(t)$$

$$c(t) = x_1$$

$$\frac{dx_1}{dt} = \frac{dc(t)}{dt} = x_2$$

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = \frac{d^2 c(t)}{dt^2} = x_3$$

$$\frac{dx_2}{dt} = x_3$$

$$\frac{d^3 x_1}{dt^3} = -9 \frac{d^2 x_1}{dt^2} - 26 \frac{dx_1}{dt} - 24x_1(t) + 24r(t)$$

$$\frac{d^3 x_1}{dt^3} = -9x_3 - 26x_2 - 24x_1 + 24r(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r(t)$$

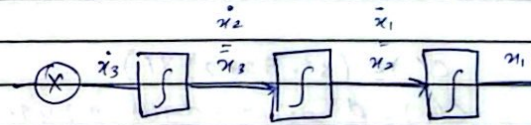
DAY-10

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

$$R(s)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

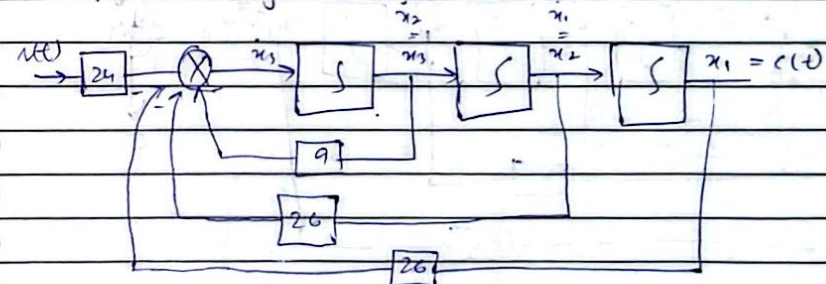
$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



→ the no. of integrators will be equal to state variable (n)

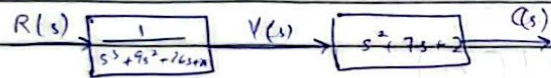
$$\frac{d^3 c}{dt^3} + 9 \frac{d^2 c}{dt^2} + 26 \frac{dc}{dt} + 24c = 24r$$

updated diagram is



TF is defined for one input & one output

$$\frac{C(s)}{R(s)} = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$



$$Y(s) = \frac{1}{s^3 + 9s^2 + 26s + 24} \{ R(s) \}$$

$$y = x_1$$

$$\frac{dy}{dt} = x_2$$

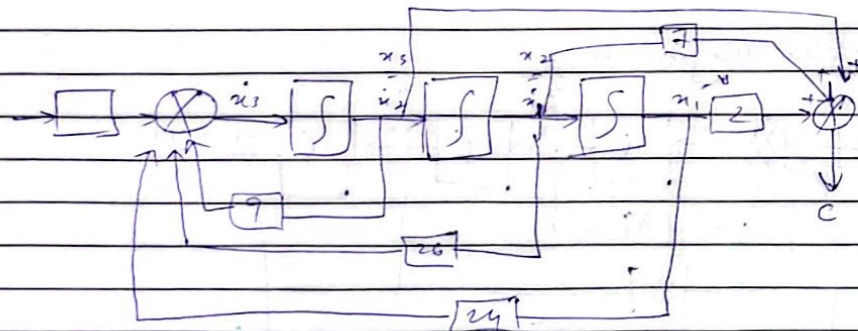
$$\frac{d^2y}{dt^2} = x_3$$

$$C(s) = (s^2 + 7s + 2) Y(s)$$

$$C(t) = \frac{d^2y(t)}{dt^2} + 7 \frac{dy(t)}{dt} + 2y(t)$$

$$C = x_3 + 7x_2 + 2x_1$$

$$C = \begin{bmatrix} 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



Converting State Space Model into TF:

$$\dot{X} = AX + BU \Rightarrow sX(s) = AX(s) + BU(s)$$

$$Y = CX + DU \Rightarrow Y(s) = CX(s) + DU(s)$$

$$sX(s) - AX(s) = BU(s)$$

$$X(s) [sI - A] = BU(s)$$

$$X(s) = [sI - A]^{-1} BU(s)$$

$$Y(s) = C [sI - A]^{-1} BU(s) + DU(s)$$

$$= [C(sI - A)^{-1}B + D] U(s)$$

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} U$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{\text{adj}(sI - A)}{\det(sI - A)}$$

$$\det(SI - A) = s(s(s+3)+2) + 1(0+1)$$

$$= s^3 + 3s^2 + 2s + 1$$

Adj A = [Matrix of cofactors]

DAY-11

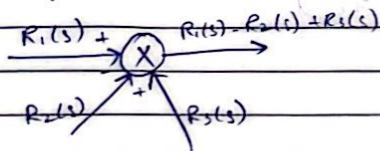
Block Diagram Reduction:

Components of a block diagram:

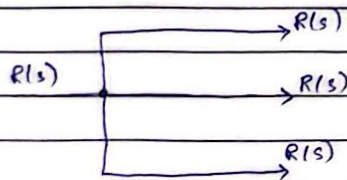
a) $C(s)$

b) $R(s) \rightarrow G(s) \rightarrow C(s)$

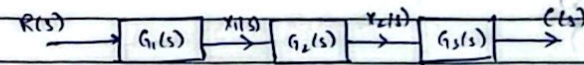
c) Summing Point:



d) Pick off Point / Branch Point:



Cascade Systems:

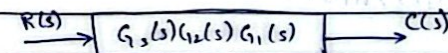


$$X_1(s) = G_1(s) R(s)$$

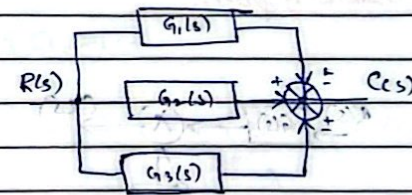
$$X_2(s) = G_2(s) X_1(s) = G_2(s) G_1(s) R(s)$$

$$C(s) = G_3(s) X_2(s)$$

$$= G_3(s) G_2(s) G_1(s) R(s)$$

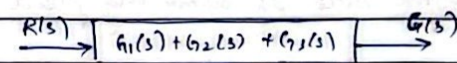


Parallel Sub-Systems:



$$C(s) = G_1(s) R(s) + G_2(s) R(s) + G_3(s) R(s)$$

$$= R(s) (G_1(s) + G_2(s) + G_3(s))$$



Feedback System:



$$C(s) = G(s)E(s) \rightarrow \textcircled{1}$$

$$E(s) = R(s) - B(s)$$

$$B(s) = H(s)C(s)$$

$$E(s) = R(s) - H(s)C(s)$$

Using $\textcircled{1}$

$$C(s) = G(s)[R(s) - H(s)C(s)]$$

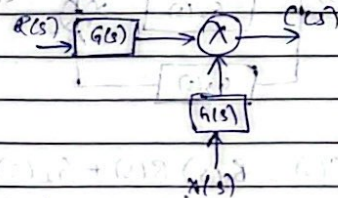
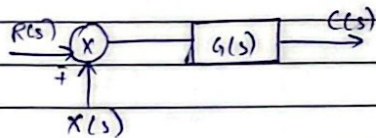
$$C(s) + G(s)H(s)C(s) = G(s)R(s)$$

$$C(s)[1 + G(s)H(s)] = G(s)R(s)$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

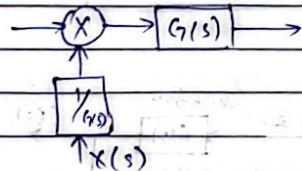
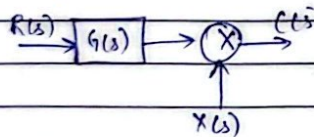
$$\frac{1}{1 + G(s)H(s)} \rightarrow \text{open loop TF}$$

Moving a TF to the left of summing point:

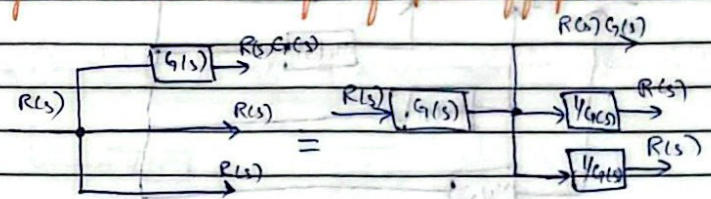


$$C(s) = G(s)[R(s) \mp X(s)]$$

Moving a block (T.F) to the right side of summing point:



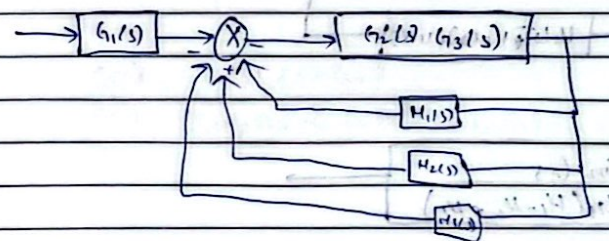
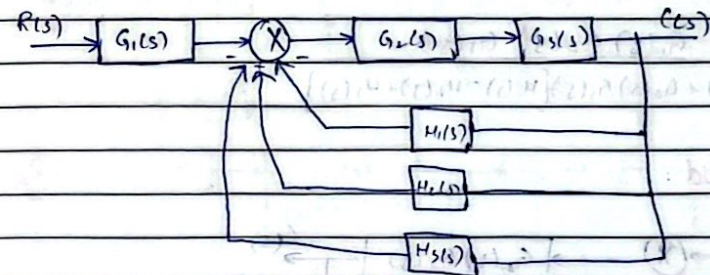
Moving a TF to the left of pick off point:

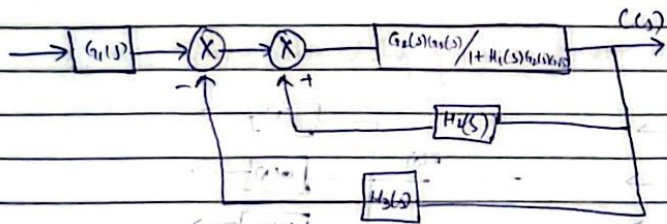
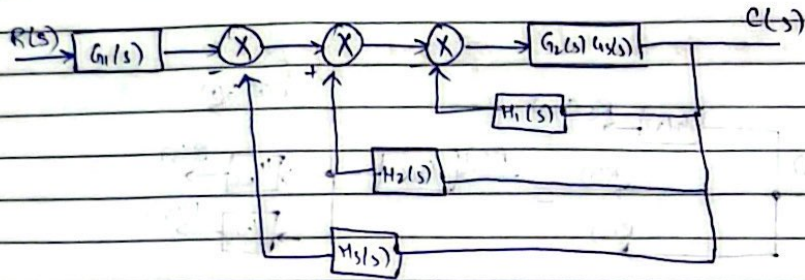


Moving a T.F to the right of pick off point:

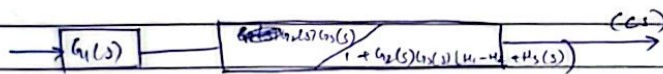


QUESTION:-



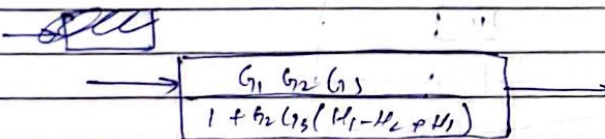
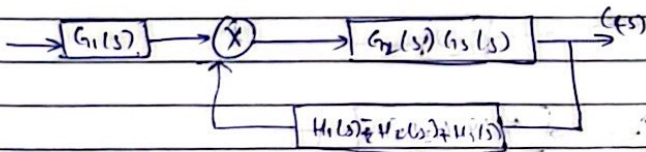


$$\frac{G(s)}{1 - G(s)H(s)}$$

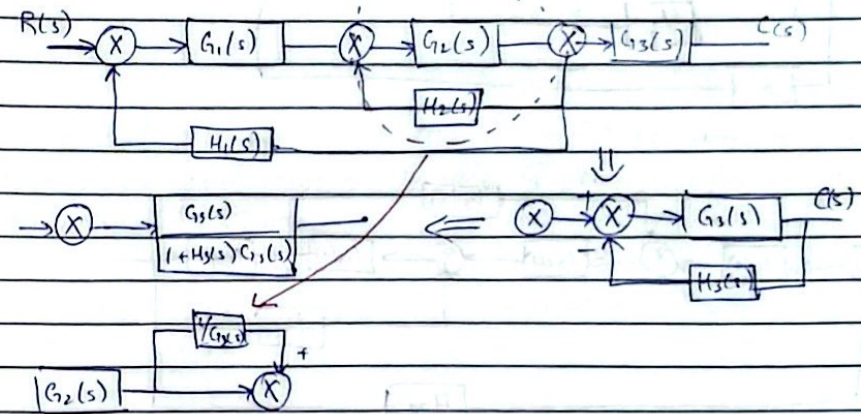


$$\frac{C(s)}{R(s)} = \frac{G_1(s) G_2(s) G_3(s)}{1 + G_2(s) G_3(s) [H_1(s) + H_2(s) + H_3(s)]}$$

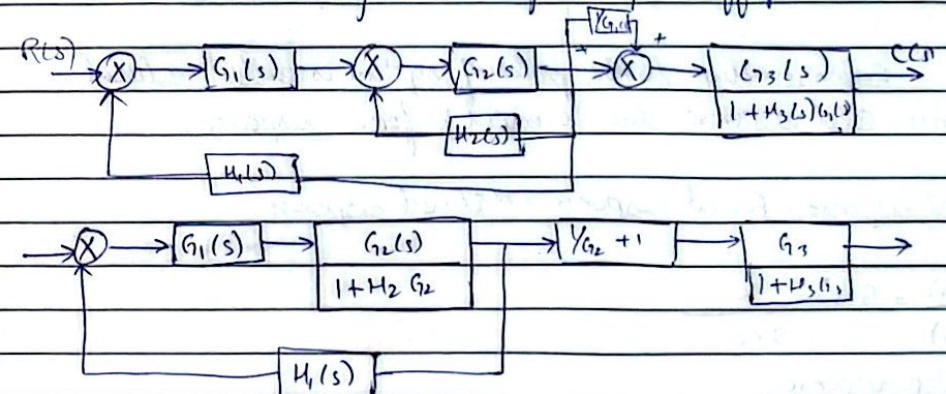
Another Method:



DAY-12

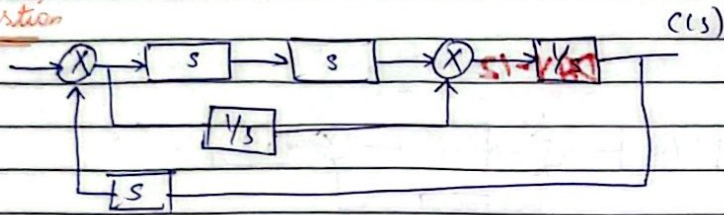


We cannot interchange summing and pick off point.

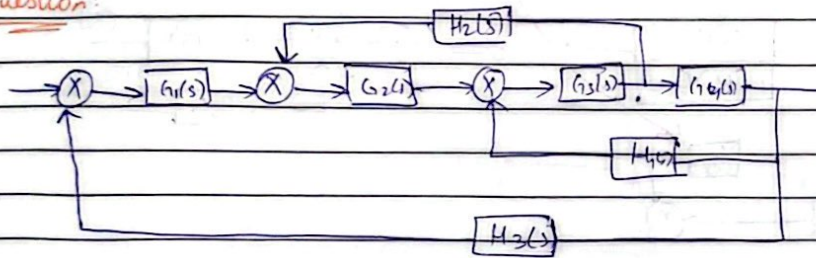


$$\frac{C(s)}{R(s)} = \frac{G_1(s) G_2(s) G_3(s)}{1 + G_2(s) G_3(s) [H_1(s) + H_2(s) + H_3(s)]}$$

Question



Question:



Ch#4 Time Analysis of First order systems

Response due to the poles of system is called natural response and response due to input is force response.

Total response = Forced response + Natural response

$$\frac{C(s)}{R(s)} = G(s) = \frac{K}{s+a}$$

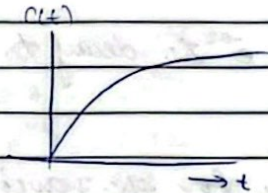
step response

$$C(s) = \frac{K}{s(s+a)}$$

$$= \frac{K/a}{s} - \frac{K/a}{s+a}$$

Forced response Natural response

$$c(t) = \frac{K}{a} (1 - e^{-at}) u(t)$$



Every real pole in the s domain introduces an exponential term in the final response. The rate of decay or fall depends on the location of pole.

$$G(s) = \frac{s+2}{s+5}$$



$$C(s) = \frac{s+2}{s(s+5)}$$

$$= \frac{A}{s} + \frac{B}{s+5}$$

$$= \frac{2/5}{s} + \frac{3/5}{s+5}$$

$$c(t) = \frac{2}{5} u(t) + \frac{3}{5} e^{-5t} u(t)$$

DAY-13

Transient Response of First order system

1. Time Constant.

Time for step response to reach 63% of its final value.

OR

Time for e^{-at} to decay to 37% of its initial value.

2. Rise time:

Time for the waveform to go from 0.1-0.9 (10-90%) of its final value.

$$Tr = t_2 - t_1$$

3. Settling time:

Time for the response to reach and stay within 2% of its final value. (98% or 0.98)

→ For time constant:

$$G(s) = \frac{K}{s+a}$$

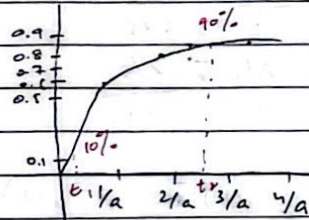
$$c(t) = \frac{K}{a} (1 - e^{-at})$$

Assume $K=a$

$$G(s) = \frac{a}{s+a}$$

$$C(s) = \frac{a}{s(s+a)}$$

$$c(t) = (1 - e^{-at}) u(t)$$



$c(t)$ at $t = 1/a$

$$c(t) = 1 - e^{-a(1/a)} = 0.632$$

81-YAQ Time constant = $1/a$

at $t = 2/a$

$$c(t) = 1 - e^{-a(2/a)} = 0.864$$

at $t = 3/a$

$$c(t) = 1 - e^{-a(3/a)} = 0.950$$

at $t = 4/a$

$$c(t) = 1 - e^{-a(4/a)} = 0.981$$

where 'a' is the location of pole.

→ For rise time:

$$c(t) = 1 - e^{-at}$$

$$0.1 = 1 - e^{-at_1}$$

$$e^{-at_1} = 1 - 0.1$$

$$-at_1 = -0.105$$

$$t_1 = 0.105$$

$$a$$

$$tr = t_2 - t_1$$

$$= 2.302 - 0.105$$

$$a$$

$$a$$

$$tr = \frac{2.2}{a}$$

$$c(t) = 1 - e^{-at}$$

$$0.9 = 1 - e^{-at_2}$$

$$e^{-at_2} = 1 - 0.9$$

$$-at_2 = -2.302$$

$$t_2 = 2.302$$

$$a$$

sabse phle iska effect sabse zaida
decay ho ga.

→ For settling time:

$$c(t) = 1 - e^{-at}$$

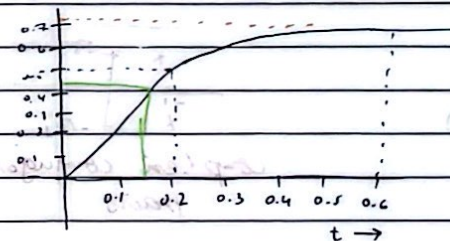
$$0.98 = 1 - e^{-at}$$

$$e^{-at} = 1 - 0.98$$

$$-at = -3.91$$

$$ts = \frac{4}{a}$$

Question :-



Final value = 0.72

$$c(t)|_{t=\infty} = 0.63 \times 0.72 = 0.45$$

$$\text{time constant} = \frac{1}{a} = 0.15 \Rightarrow a = 6.66$$

$$c(t) = \frac{k}{a} (1 - e^{-at})$$

$$\frac{k}{a} = \text{final value}$$

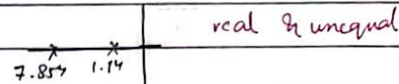
$$\frac{k}{a} = 0.72$$

$$\begin{aligned} k &= 0.72 \times a \\ &= 0.72 \times 6.6 \\ &= 4.75 \end{aligned}$$

$$G(s) = \frac{4.75}{s + 6.6}$$

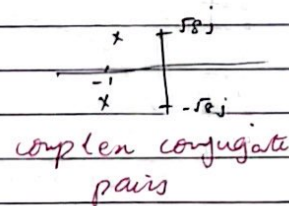
$$G_1(s) = \frac{9}{s^2 + 9s + 9}$$

$$= \frac{9}{(s + 7.854)(s + 1.146)}$$



$$G_2(s) = \frac{9}{s^2 + 2s + 9}$$

$$= \frac{9}{(s + 1 + j\sqrt{8})(s + 1 - j\sqrt{8})}$$



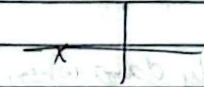
$$G_3(s) = \frac{9}{s^2 + 9}$$

$$= \frac{9}{(s + 3j)(s - 3j)}$$



$$G_4(s) = \frac{9}{s^2 + 6s + 9}$$

$$= \frac{9}{(s + 3)^2}$$

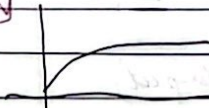
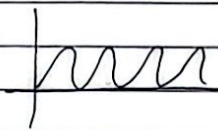
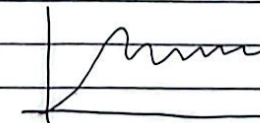


$$c_1(t) = 1 + 0.171 e^{-7.854t} - 1.171 e^{-1.146t}$$

$$c_2(t) = 1 - 1.06 e^{-t} \cos(\sqrt{8}t - 19.47^\circ)$$

$$c_3(t) = 1 - \cos 3t$$

$$c_4(t) = 1 - 3t e^{-3t} - e^{-3t}$$



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$\omega_n \rightarrow$ natural frequency
 $\zeta \rightarrow$ damping ratio

critically damp when poles are equal and if not equal, they are over damp

DAY-14

Second Order System:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$\omega_n =$ natural frequency

$\zeta =$ Damping ratio

$\zeta = 0 \Rightarrow$ Undamped system

$0 < \zeta < 1 \Rightarrow$ Underdamped system

$\zeta = 1 \Rightarrow$ critically damped system

$\zeta > 1 \Rightarrow$ Overdamped system

$$\frac{C(s)}{R(s)} = G_1(s) = \frac{9}{s^2 + 9s + 9}$$

$$R(s) = 1/s$$

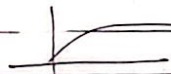
$$r(t) = u(t)$$

$$C(s) = \frac{9}{s(s + 7.854)(s + 1.146)}$$

$$c(t) = 1 + 0.17e^{-7.854t} - 1.171e^{-1.146t}$$

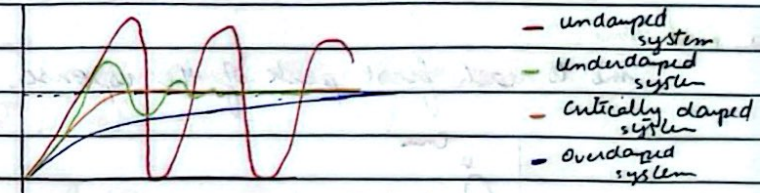
$$\omega_n = 9 \Rightarrow \omega_n = 3$$

$$2\zeta\omega_n = 9 \Rightarrow \zeta = 9/6 \Rightarrow 1.5 > 1 \text{ (overdamped)}$$



$$\rightarrow G_2(s) = \frac{9}{s^2 + 2s + 9}$$

$$c(t) = 1 - 1.06e^{-t} \cos(\sqrt{8}t - 19.47^\circ)$$



$$\omega_n = 3$$

$$2\zeta\omega_n = 2$$

$$\zeta = \frac{1}{3}$$

$0 < \zeta < 1 \Rightarrow$ Underdamped.

$$\rightarrow G_3(s) = \frac{9}{s^2 + 9}$$

$$c(t) = 1 - \cos 3t$$

$$\omega_n = 3$$

$$2\zeta\omega_n = 0$$

$\zeta = 0 \Rightarrow$ undamped

$$\rightarrow G_4(s) = \frac{9}{(s+3)^2} \Rightarrow \frac{9}{s^2 + 6s + 9}$$

$$\omega_n = 3$$

$$2\zeta\omega_n = 6$$

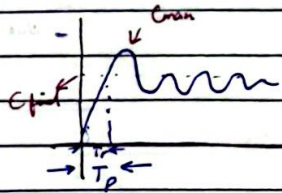
$\zeta = 1 \Rightarrow$ critically damped

Transient Response of underdamped second order system:1. Rise time: reach waveform

Time to 10-90% of the final value.

2. Peak time:

Time to reach first peak of the response.

Maximum Overshoot:

The amount that the waveform overshoots the steady state or final value at the peak time expressed as a percentage of steady state value.

$$\%OS = \frac{C_{max} - C_{final}}{C_{final}} \times 100$$

Response of second order system:

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$A = 1, B = -1, C = -2\zeta\omega_n$$

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

$$= \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2} + \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2}$$

$$c(t) = 1 - \left[e^{-\zeta\omega_n t} \cos(\omega_n\sqrt{1 - \zeta^2} t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n\sqrt{1 - \zeta^2} t) \right]$$

$$= 1 - e^{-\zeta\omega_n t} \left[\frac{\cos(\omega_n\sqrt{1 - \zeta^2} t + \phi)}{\sqrt{1 - \zeta^2}} \right]$$

$$\cos A \cos B - \sin A \sin B = \cos(A + B)$$

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \cos(\omega_n\sqrt{1 - \zeta^2} t + \phi)$$

$$\phi = \tan^{-1} \frac{\zeta}{\sqrt{1 - \zeta^2}}$$

DAY-15

Step Response of Second Order System:

$$c(t) = e^{-\zeta\omega_n t} \left(\cos \omega_n\sqrt{1 - \zeta^2} t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_n\sqrt{1 - \zeta^2} t \right)$$

$$= 1 - e^{-\zeta\omega_n t} \cos(\omega_n\sqrt{1 - \zeta^2} t + \phi)$$

$$\phi = \tan^{-1} \frac{\zeta}{\sqrt{1 - \zeta^2}}$$



Tabletop

Date:

Sun Mon Tue Wed Thu Fri Sat

Peak time = T_p

$$\frac{dC(t)}{dt} \leftrightarrow s C(s)$$

$$s C(s) = \frac{d}{dt} [C(t)] = s \cdot \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

$$\begin{aligned} e^{at} \sin \omega t &\leftrightarrow \frac{\omega}{(s+a)^2 + \omega^2} \\ e^{-at} \cos \omega t &\leftrightarrow \frac{s+a}{(s+a)^2 + \omega^2} \end{aligned}$$

$$\frac{dC(t)}{dt} = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2} t = 0$$

$$\begin{aligned} \sin \omega_n \sqrt{1-\zeta^2} t &= 0 \\ \omega_n \sqrt{1-\zeta^2} t &= t\pi \\ \omega_n \sqrt{1-\zeta^2} T_p &= \pi \end{aligned}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

Max Overshoot H_p
% Os :

$C_{max} - C_{final}$
 C_{final}

Date:

Sun Mon Tue Wed Thu Fri Sat

$$C(t) = 1 - e^{-\zeta\omega_n t} \left(\cos(\omega_n \sqrt{1-\zeta^2} t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_n \sqrt{1-\zeta^2} t \right)$$

$$C_{max} = C(T_p) \quad T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$C(t) = 1 - e^{-\frac{\zeta\omega_n \pi}{\omega_n \sqrt{1-\zeta^2}}} \left(\cos \omega_n \sqrt{1-\zeta^2} \cdot \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_n \sqrt{1-\zeta^2} \cdot \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \right)$$

$$\begin{aligned} C_{max} &= 1 + e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \\ \% Os &= e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 \end{aligned}$$

Settling Time

$$T_s = \frac{4}{2\zeta\omega_n}$$

31-VAG

Location of Poles of second order system in terms of ζ & ω_n

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

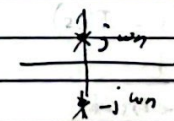
$$s = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$= \frac{-2\zeta\omega_n \pm 2\omega_n \sqrt{\zeta^2 - 1}}{2}$$

$$= -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

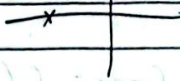
$\zeta = 0$ underdamped

$$s = \pm j\omega_n$$

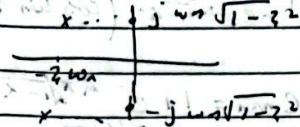


COPY

$\zeta = 1$ critically damped
 $s = -\omega_n \pm 0$

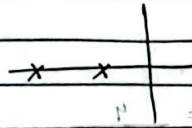


$0 < \zeta < 1$ underdamped
 $s = -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2}$



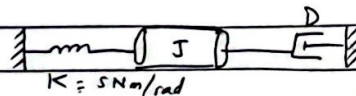
$\zeta > 1$ overdamped

$$s = -\zeta \omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$



DAY-16

Question:-



Find J and D to yield 20% overshoot and $T_s = 2$ sec

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d}$$

$$T_s = \frac{4}{\zeta \omega_n}$$

$$\%Os = e^{-\frac{\pi \zeta}{\sqrt{1 - \zeta^2}}} \times 100 \rightarrow \textcircled{1}$$

$$(J s^2 + D s + K) \theta(s) = T(s)$$

$$\theta(s) = \frac{1}{T(s)}$$

$$T(s) = J s^2 + D s + K$$

E1-YAG 20

$$= \frac{1/J}{s^2 + D/J s + K/J}$$

$$\omega_n^2 = \frac{K}{J}$$

$$\omega_n = \sqrt{\frac{K}{J}}$$

$$2 \zeta \omega_n = \frac{D}{J}$$

Using ①

$$0.2 = e^{-\frac{\pi \zeta}{\sqrt{1 - \zeta^2}}} \times 100$$

$$\zeta = -\frac{\ln(\%Os)}{\sqrt{\pi^2 + \ln^2(\%Os/100)}}$$

$$\zeta = 0.456$$

$$T_s = 2 \text{ sec}$$

$$T_s = 4$$

$$2 \omega_n$$

$$\omega_n = 4.4$$

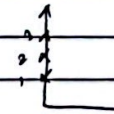
$$J = 0.26 \text{ kg/m}^2$$

$$D = 0.3 \text{ Nm/rad}$$

Relationship b/w damping ratio & percent overshoot

Damping ratio and %Os $\rightarrow$ Inverse

Damping ratio and risetime $\rightarrow$ Direct



1 ka peaktime sabse bada ho ga.

Date:.....

Sun Mon Tue Wed Thu Fri Sat

Oh DAY-17

Requirements on the design of control systems

- Transient response
- stability
- Steady state error

Baki slides se parhna haiiii